

# Caelum Cromwell

**Contact Information** | Email: caelum.cromwell@email.com | Phone: +44 7700 900123 | LinkedIn: linkedin.com/in/caelumcromwell | GitHub: github.com/caelumcromwell

## Professional Summary

Highly motivated and results-oriented Computer Vision Specialist with 5+ years of experience at Synthetica Global, specializing in developing and deploying cutting-edge AI/ML solutions. Proven ability to design, implement, and optimize complex algorithms for image recognition, object detection, and image segmentation. Proficient in various programming languages (C++, Python, R), deep learning frameworks (PyTorch, TensorFlow), and cloud platforms (GCP). Seeking a challenging role where I can leverage my expertise to contribute to innovative projects and drive significant impact.

## Education

**Imperial College London**, London, UK \* **Master of Science (MSc), Computer Science**, 2018 \* Specialization: Artificial Intelligence \* GPA: 3.9/4.0 \* Thesis: "Real-time Object Tracking using Deep Convolutional Neural Networks and Kalman Filtering"

**University of Oxford**, Oxford, UK \* **Bachelor of Science (BSc), Mathematics**, 2016 \* GPA: 3.7/4.0

## Technical Skills

**Programming Languages:** C++, Python, R, Java (Intermediate)

**Deep Learning Frameworks:** PyTorch, TensorFlow, Keras

**Machine Learning Libraries:** Scikit-learn, XGBoost, LightGBM

**Cloud Platforms:** Google Cloud Platform (GCP) – Compute Engine, Cloud Storage, Cloud Functions, BigQuery

**Databases:** SQL, NoSQL (MongoDB)

**Computer Vision Libraries:** OpenCV, scikit-image

**Other:** Docker, Kubernetes, Git, Agile methodologies

# Professional Experience

**Synthetica Global**, London, UK **Computer Vision Specialist** | June 2018 – Present

- Developed and deployed a real-time object detection system for autonomous vehicles using YOLOv5, achieving 95% accuracy and a processing speed of 30 FPS.
- Designed and implemented a deep learning model for image segmentation to identify defects in manufactured products, resulting in a 15% reduction in production errors.
- Led a team of three engineers in the development of a facial recognition system for security applications, integrating the system with existing infrastructure and achieving 98% accuracy.
- Contributed to the development of a novel approach to image super-resolution using generative adversarial networks (GANs), resulting in a publication in the International Conference on Computer Vision (ICCV).
- Mentored junior engineers in best practices for software development and AI/ML model deployment.
- Regularly presented project updates and findings to senior management.

## AI/ML Projects

### Real-time Object Tracking System (Personal Project)

- Developed a real-time object tracking system using OpenCV and deep learning, capable of tracking multiple objects simultaneously.
- Utilized Kalman filtering to improve the accuracy and robustness of the tracking algorithm.
- Achieved an average tracking accuracy of 92% on the MOT17 benchmark dataset.

### Image Classification using Transfer Learning (Personal Project)

- Implemented an image classification model using transfer learning with ResNet50, achieving state-of-the-art results on the CIFAR-10 dataset.
- Explored different data augmentation techniques to improve model performance and robustness.
- Achieved 98% accuracy on the test set.

### Predictive Maintenance using Time Series Analysis (Synthetica Global)

- Developed a predictive maintenance model for industrial machinery using time series analysis and machine learning algorithms.

- Utilized LSTM networks to predict equipment failures with a lead time of 7 days, resulting in a 10% reduction in unplanned downtime.
- Deployed the model to a GCP cloud environment for real-time monitoring and prediction.